

Linear Algebra 2

Angus

2026.10.1

Abstract

1 Lecture 1

2 Lecture 2

3 Lecture 3

4 Lecture 4

5 Lecture 5

6 Lecture 6

How to relate different vector spaces.

6.1 Linear maps

Definition 6.1

Linear map

Let V and W be vector spaces over a field $\mathbb{F}$. A map $T : V \rightarrow W$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$T(u + v) = T(u) + T(v), \quad T(\alpha v) = \alpha T(v).$$

Additivity and homogeneity

- Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$.
- Homogeneity: $T(\alpha v) = \alpha T(v)$ for all $\alpha \in \mathbb{F}, v \in V$.

Notation

- $\mathcal{L}(V, W)$ = the space of linear maps $T : V \rightarrow W$
- $\mathcal{L}(V)$ = the space of linear maps $T : V \rightarrow V$

6.2 Examples

6.2.1 The zero linear map

$$\begin{aligned} 0 : V &\rightarrow W, \\ v &\mapsto 0. \end{aligned}$$

Remark

$$T(0_V) = 0_W.$$

$$\begin{aligned} 0_V &= 0_{\mathbb{F}}v, \\ T(0_V) &= T(0_{\mathbb{F}}v) = 0_{\mathbb{F}}T(v) = 0_W. \end{aligned}$$

6.2.2 The identity linear map

For any $v \in V$,

$$\begin{aligned} I : V &\rightarrow V, \\ v &\mapsto v. \end{aligned}$$

6.2.3 Differentiation operator

First recall the definition of a polynomial space over $\mathbb{R}$.

$$\mathcal{P}(\mathbb{R}) = \{\text{polynomials } a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in \mathbb{R}, n \in \mathbb{N}\}$$

Then define the differentiation operator.

$$\begin{aligned} D : \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \psi' \end{aligned}$$

For $\psi \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} \psi &= a_0 + a_1x + a_2x^2 + \cdots + a_nx^n, \\ \psi' &= a_1 + 2a_2x + \cdots + na_nx^{n-1}. \end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned} (p + q)' &= p' + q', \\ (\alpha p)' &= \alpha p'. \end{aligned}$$

The differentiation operator is linear.

6.2.4 Integration operator

Define the integration operator.

$$\begin{aligned}\int : \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto \int \psi \, dx\end{aligned}$$

For all $\alpha \in \mathbb{R}$ and all polynomials $p, q \in \mathcal{P}(\mathbb{R})$,

$$\begin{aligned}\int (p + q) &= \int p + \int q, \\ \int (\alpha p) &= \alpha \int p.\end{aligned}$$

6.2.5 Multiplication by a fixed polynomial

Define the multiplication by a fixed polynomial operator.

$$\begin{aligned}T : \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ \psi &\mapsto q\psi\end{aligned}$$

So

$$(Tp)(x) = q(x)p(x).$$

Check that T is linear.

6.2.6 Backward shift operator

Define the backward shift operator.

$$\begin{aligned}T : \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (x_2, x_3, \dots)\end{aligned}$$

Check that T is linear.

6.2.7 Forward shift operator

Define the forward shift operator.

$$\begin{aligned}T : \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty \\ (x_1, x_2, \dots) &\mapsto (0, x_1, x_2, \dots)\end{aligned}$$

Check that T is linear.

6.2.8 Linear map and matrices

$$\begin{aligned}T : \mathbb{F}^n &\rightarrow \mathbb{F}^m, \\ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} &\mapsto \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}.\end{aligned}$$

T should be a matrix.

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{(m \times n)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$m \times n$ matrix: rows $\times$ columns.

$$T \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, T \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, T \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Columns!

So, because the lemma below shows that a linear map is completely determined by its action on the basis vectors of the domain. Any linear map can be uniquely represented by a matrix with respect to the chosen bases.

6.3 A linear map is determined by its values on a basis

Lemma 6.1

A linear map $T: V \rightarrow W$ is uniquely determined by what it does on a basis of V .

Example

For a basis $\{v_1, \dots, v_n\}$ of V , let

$$T(v_i) = w_i \in W \quad \text{for } i = 1, 2, \dots, n.$$

$$T(v_1) = w_1,$$

$$T(v_2) = w_2,$$

$$\vdots$$

$$T(v_n) = w_n.$$

Then T is uniquely determined by the images of the basis vectors.

$$\begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix}$$

determines T uniquely. For $v \in V$,

$$\begin{aligned} Tv &= T(a_1v_1 + a_2v_2 + \dots + a_nv_n) \\ &= a_1T(v_1) + a_2T(v_2) + \dots + a_nT(v_n) \\ &= a_1w_1 + a_2w_2 + \dots + a_nw_n. \end{aligned}$$

Check that T is a linear map.

6.3.1 $\mathcal{L}(V, W)$

Let $S, T \in \mathcal{L}(V, W)$.

$$\begin{aligned} (S + T)(v) &= S(v) + T(v) && \text{for all } v \in V, \\ (\lambda S)(v) &= \lambda S(v) && \text{for all } v \in V \text{ and } \lambda \in \mathbb{F}. \end{aligned}$$

Claim (Need to check)

$S + T, \lambda S$ are linear maps.

That makes $\mathcal{L}(V, W)$ a vector space over $\mathbb{F}$.

Similarly, $\mathcal{L}(V)$ is a vector space over $\mathbb{F}$ for any vector space V .

6.3.2 Compositions of linear maps

Case 1 Let U, V, W be vector spaces over $\mathbb{F}$.

$$\begin{array}{ccccccc} & & & & \text{\scriptsize } R \circ T & & \\ & & & \nearrow & & \searrow & \\ U & \xrightarrow{S} & V & \xrightarrow{T} & W & \xrightarrow{R} & X \\ & \searrow & & \nwarrow & & \nearrow & \\ & & & & \text{\scriptsize } T \circ S & & \end{array}$$

For all $u \in U$,

$$\begin{aligned} (T \circ S)(u) &= T(S(u)), \\ (R \circ (T \circ S))(u) &= R(T(S(u))), \\ ((R \circ T) \circ S)(u) &= (R \circ T)(S(u)) = R(T(S(u))). \end{aligned}$$

Therefore,

$$R \circ (T \circ S) = (R \circ T) \circ S.$$

Case 2 For $T: V \rightarrow W$.

$$\begin{array}{ccccccc} V & \xrightarrow{I_V} & V & \xrightarrow{T} & W & \xrightarrow{I_W} & W \\ & \searrow & & \nwarrow & & \nearrow & \\ & & & & T & & \end{array}$$

So

$$\begin{aligned} T \circ I_V &= T, \\ I_W \circ T &= T. \end{aligned}$$

For $S, T: V \rightarrow V$, the compositions $S \circ T$ and $T \circ S$ need not agree.

Example

For $V = \mathcal{P}(\mathbb{R})$,

$$T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$p(x) \mapsto p'(x)$$

$$S: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$p(x) \mapsto xp(x)$$

$$T(p(x)) = p'(x), S(p(x)) = xp(x).$$

$$(S \circ T)(p(x)) = S(p'(x)) = xp'(x),$$

$$(T \circ S)(p(x)) = T(xp(x)) = p(x) + xp'(x).$$

6.4 Kernel and Image of a linear map

Definition 6.2

Kernel and Image Spaces

Let $T: V \rightarrow W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\} \quad \text{kernel of } T \text{ (null space of } T),$$

$$\text{Im } T = \{T(v) \mid v \in V\} \quad \text{image of } T \text{ (range of } T).$$

Example

Zero map For the zero map,

$$0: V \rightarrow W$$

$$v \mapsto 0$$

We can immediately see that the range of the zero map is just the zero vector in W , and the null space is the entire space V .

$$\text{Im}(0) = \{0\},$$

$$\ker(0) = V.$$

Example

Differentiation map For the differentiation map,

$$D: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$$

$$\gamma \mapsto \gamma'$$

We can immediately see that the range of the differentiation map is the set of all derivatives of functions in $\mathcal{P}(\mathbb{R})$, and the null space is the set of all constant polynomials.

$$\text{Im}(D) = \{\gamma' \mid \gamma \in \mathcal{P}(\mathbb{R})\},$$

$$\ker(D) = \{\text{constant polynomials}\}.$$

7 Lecture 7

7.1 Properties of Linear Maps

Let V and W be vector spaces over the same field $\mathbb{F}$. Let $T: V \rightarrow W$ be a linear map. For all $v_1, v_2 \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{cases} T(v_1 + v_2) = T(v_1) + T(v_2), \\ T(\alpha v_1) = \alpha T(v_1), \end{cases} \iff T(u + \alpha v) = T(u) + T(\alpha v) = T(u) + \alpha T(v).$$

7.2 Injective and Surjective Maps

7.2.1 Kernel and Image Definition

Definition 7.1

The kernel of T

Let $T: V \rightarrow W$ be a linear map.

$$\ker T = \{v \in V \mid T(v) = 0_W\}.$$

Definition 7.2

The image of T

Let $T: V \rightarrow W$ be a linear map.

$$\operatorname{Im} T = \{T(v) \mid v \in V\}.$$

Theorem 7.1

The kernel and image of a linear map T are vector subspaces of V and W , respectively.

$$\ker T \subseteq V,$$

$$\operatorname{Im} T \subseteq W.$$

Proof. Since $T(0_V) = 0_W$, the kernel contains 0_V .

If $v_1, v_2 \in \ker T$ and $\lambda \in \mathbb{F}$, then

$$T(v_1 + \lambda v_2) = T(v_1) + \lambda T(v_2) = 0_W,$$

so $v_1 + \lambda v_2 \in \ker T$.

Thus $\ker T$ is a subspace of V . □

Proof. If $w_1, w_2 \in \operatorname{Im} T$, write

$$w_1 = T(v_1),$$

$$w_2 = T(v_2)$$

for some $v_1, v_2 \in V$. For $\lambda \in \mathbb{F}$,

$$w_1 + \lambda w_2 = T(v_1) + \lambda T(v_2) = T(v_1 + \lambda v_2) \in \operatorname{Im} T.$$

Thus $\operatorname{Im} T$ is a subspace of W . □

7.2.2 Injective

Definition 7.3

The linear map $T: V \rightarrow W$ is called *injective* (or one-to-one) if $\ker T = \{0_V\}$.

For $T: V \rightarrow W$, injectivity is equivalent to

$$T(v_1) = T(v_2) \implies v_1 = v_2.$$

Indeed, if $T(v_1) = T(v_2)$, then

$$T(v_1 - v_2) = 0_W,$$

so

$$v_1 - v_2 \in \ker T = \{0_V\},$$

and hence

$$v_1 = v_2.$$

Conversely, if T is injective and $v \in \ker T$, then

$$T(v) = T(0_V),$$

so $v = 0_V$.

7.2.3 Surjective

Definition 7.4

The linear map $T: V \rightarrow W$ is called *surjective* (or onto) if $\operatorname{Im} T = W$.

If T is surjective, then by definition $\operatorname{Im} T = W$.

Equivalently, for every $w \in W$ there exists $v \in V$ such that

$$w = T(v).$$

Example

The derivative map

$$\begin{aligned} D: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}) \\ p &\mapsto p' \end{aligned}$$

has

$$\begin{aligned} \ker D &= \{p \in \mathcal{P}(\mathbb{R}) \mid D(p) = 0\}, \\ \operatorname{Im} D &= \{D(p) \mid p \in \mathcal{P}(\mathbb{R})\}. \end{aligned}$$

D is surjective but not injective.

Example

The backward shift b and forward shift f on $\mathbb{F}^\infty$ are defined by

$$\mathbb{F}^\infty \begin{matrix} \xrightarrow{b} \\ \xrightarrow{f} \end{matrix} \mathbb{F}^\infty$$

$$\begin{aligned} b(x_1, x_2, x_3, \dots) &= (x_2, x_3, x_4, \dots), \\ f(x_1, x_2, x_3, \dots) &= (0, x_1, x_2, \dots). \end{aligned}$$

The kernels and images of the backward and forward shifts are given by

$$\begin{aligned} \ker b &= \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_2 = x_3 = \dots = 0\}, \\ \operatorname{Im} b &= \mathbb{F}^\infty, \\ \ker f &= \{0\}, \\ \operatorname{Im} f &= \{(x_1, x_2, x_3, \dots) \in \mathbb{F}^\infty \mid x_1 = 0\}. \end{aligned}$$

The backward shift is surjective but not injective, while the forward shift is injective but not surjective.

Example

$$\begin{aligned} T: \mathcal{P}(\mathbb{R}) &\rightarrow \mathcal{P}(\mathbb{R}), \\ p(x) &\mapsto xp(x). \end{aligned}$$

$$\ker T = \{p(x) \in \mathcal{P}(\mathbb{R}) \mid xp(x) = 0\} = \{0\}.$$

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n, \quad a_i \in \mathbb{R}.$$

$$xp(x) = a_0x + a_1x^2 + a_2x^3 + \dots + a_nx^{n+1}.$$

$$\operatorname{Im} T = \{xp(x) \mid p(x) \in \mathcal{P}(\mathbb{R})\} \neq \mathcal{P}(\mathbb{R}).$$

T is injective but not surjective.

7.3 The Fundamental Theorem of Linear maps

Theorem 7.2

The fundamental theorem of linear maps (also known as the rank-nullity or rank theorem)

If $T: V \rightarrow W$ is a linear map, and the dimension of V is finite, then

$$\dim V = \underbrace{\dim \ker T}_{\text{nullity}} + \underbrace{\dim \operatorname{Im} T}_{\text{rank}}.$$

Proof. Since $\ker T \subseteq V$, let $\{u_1, \dots, u_n\}$ be a basis for $\ker T$. Extend it to a basis for V .

$$\{u_1, \dots, u_n, v_1, \dots, v_r\}$$

Thus

$$\dim V = n + r, \quad \dim \ker T = n.$$

We want to show that $\dim \operatorname{Im} T = r$, so we claim that $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{Im} T$, need to be proven.

Claim: $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

1. $\{T(v_1), \dots, T(v_r)\}$ spans $\text{Im } T$. Since $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V , any $v \in V$ can be written as

$$v = a_1 u_1 + \dots + a_n u_n + b_1 v_1 + \dots + b_r v_r.$$

Applying T , we get

$$T(v) = b_1 T(v_1) + \dots + b_r T(v_r),$$

since $T(u_i) = 0$ for all i . Therefore, $\{T(v_1), \dots, T(v_r)\}$ spans $\text{Im } T$.

2. $\{T(v_1), \dots, T(v_r)\}$ is linearly independent. Suppose

$$c_1 T(v_1) + \dots + c_r T(v_r) = 0.$$

Then

$$T(c_1 v_1 + \dots + c_r v_r) = 0.$$

Therefore, $c_1 v_1 + \dots + c_r v_r \in \ker T$. But $\{u_1, \dots, u_n, v_1, \dots, v_r\}$ is a basis for V , so $c_1 v_1 + \dots + c_r v_r$ can be written uniquely as a linear combination of the basis vectors:

$$c_1 v_1 + \dots + c_r v_r = a_1 u_1 + \dots + a_n u_n.$$

Hence

$$c_1 v_1 + \dots + c_r v_r - (a_1 u_1 + \dots + a_n u_n) = 0.$$

Since the basis vectors are linearly independent, we must have

$$c_1 = \dots = c_r = 0.$$

This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim. Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

$$\dim \text{Im } T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

$$a_1 = \dots = a_n = 0, \quad c_1 = \dots = c_r = 0.$$

This proves that $\{T(v_1), \dots, T(v_r)\}$ is linearly independent, completing the proof of the claim.

Therefore, $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\text{Im } T$.

$$\dim \text{Im } T = r, \quad \dim \ker T = n, \quad \dim V = n + r.$$

□

7.4 Corollaries of the Rank Theorem

Let $T: V \rightarrow W$ with

1. $\dim V < \infty$
2. $\dim W < \infty$.

7.4.1 Injective Maps

If T is injective, then

$$\dim V = \dim \operatorname{Im} T \leq \dim W,$$

because $\ker T = \{0_V\}$.

So if $\dim V > \dim W$, no injective map $T: V \rightarrow W$ exists.

7.4.2 Surjective Maps

If T is surjective, then

$$\dim W = \dim \operatorname{Im} T \leq \dim V,$$

because $\operatorname{Im} T = W$.

So if $\dim V < \dim W$, no surjective map $T: V \rightarrow W$ exists.

7.4.3 Bijective Maps

If $T: V \rightarrow W$ is injective and surjective, then $\dim V = \dim W$.

Proof. Since T is injective,

$$\dim V = \dim \operatorname{Im} T \leq \dim W.$$

Since T is surjective,

$$\dim W = \dim \operatorname{Im} T \leq \dim V.$$

Therefore,

$$\dim V = \dim W.$$

□

8 Lecture 8

8.1 Review

For $T: V \rightarrow W$,

$$\ker T = \{v \in V \mid T(v) = 0_W\} \subseteq V,$$

$$\operatorname{Im} T = \{T(v) \mid v \in V\} \subseteq W.$$

- T is injective if $\ker T = \{0_V\}$.
- T is surjective if $\operatorname{Im} T = W$.

8.2 Invertible Maps

Definition 8.1

A linear map $T: V \rightarrow W$ is *invertible* if there exists a linear map $S: W \rightarrow V$ such that

$$S \circ T = I_V \quad \text{and} \quad T \circ S = I_W.$$

S is an *inverse* of T .

Theorem 8.1

An invertible linear map $T: V \rightarrow W$ has a unique inverse.

Proof. Suppose T has two inverses S_1 and S_2 . Then

$$T \circ S_1 = I_W \quad \text{and} \quad T \circ S_2 = I_W,$$

$$S_1 \circ T = I_V \quad \text{and} \quad S_2 \circ T = I_V.$$

Therefore,

$$S_1 = S_1 \circ I_W = S_1 \circ (T \circ S_2) = (S_1 \circ T) \circ S_2 = I_V \circ S_2 = S_2.$$

Hence $S_1 = S_2$, proving the uniqueness of the inverse. □

Theorem 8.2

If $T: V \rightarrow W$ is a linear map, then

$$T \text{ is invertible} \iff T \text{ is both injective and surjective.}$$

Proof. Proof in both directions.

1. ($\Rightarrow$) If T is invertible, then it is both injective and surjective.

Show T is injective

Let $v \in V$ such that $T(v) = 0_W$, then

$$v = T^{-1}T(v) = T^{-1}(0_W) = 0_V.$$

Thus

$$\ker T = \{0_V\} \implies T \text{ is injective.}$$

Show T is surjective

Let $w \in W$, then

$$w = T(T^{-1}(w)) \implies w \in \text{im } T.$$

So

$$W \subseteq \text{im } T \implies \text{im } T = W \implies T \text{ is surjective.}$$

2. ($\Leftarrow$) Conversely, if T is both injective and surjective, then it is invertible.

Since T is surjective, $\forall w \in W, \exists v \in V$ such that $T(v) = w$.

Define

$$\begin{aligned} T^{-1}: W &\rightarrow V, \\ w &\mapsto v \end{aligned}$$

This is well-defined because T is injective.

Then

$$T^{-1} \circ T(v) = v \quad \text{and} \quad T \circ T^{-1}(w) = T(v) = w$$

so

$$T^{-1} \circ T = I_V \quad \text{and} \quad T \circ T^{-1} = I_W$$

so T is invertible. □

Claim T^{-1} is linear. Let $w_1, w_2 \in W$ and $\alpha, \beta \in \mathbb{F}$. Then

$$\begin{aligned} T^{-1}(\alpha w_1 + \beta w_2) &= T^{-1}(\alpha T(v_1) + \beta T(v_2)) \\ &= T^{-1}(T(\alpha v_1 + \beta v_2)) \\ &= \alpha v_1 + \beta v_2 \\ &= \alpha T^{-1}(w_1) + \beta T^{-1}(w_2) \end{aligned}$$

where $v_1 = T^{-1}(w_1)$ and $v_2 = T^{-1}(w_2)$. Hence T^{-1} is linear.

Example

Consider the forward and backward linear maps

$$\begin{aligned} b: \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty, \\ (x_1, x_2, \dots) &\mapsto (x_2, x_3, \dots) \\ f: \mathbb{F}^\infty &\rightarrow \mathbb{F}^\infty, \\ (x_1, x_2, \dots) &\mapsto (0, x_1, x_2, \dots) \end{aligned}$$

b is surjective but not injective, f is injective but not surjective $\implies$ neither b nor f is invertible.

$$b \circ f = I_{\mathbb{F}^\infty} \quad \text{but} \quad f \circ b \neq I_{\mathbb{F}^\infty}.$$

This cannot happen for finite-dimensional vector spaces.

Theorem 8.3

Let $T: V \rightarrow W$ be a linear map with $\dim V = \dim W < \infty$. Then

$$T \text{ is injective} \iff T \text{ is surjective} \iff T \text{ is invertible.}$$

Proof. By the rank-nullity theorem, we have

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim W.$$

$$\begin{aligned} T \text{ injective} &\implies \dim \ker T = 0 \\ &\implies \dim \operatorname{Im} T = \dim V = \dim W \\ &\implies T \text{ is surjective.} \end{aligned}$$

$$\begin{aligned} T \text{ surjective} &\implies \dim \operatorname{Im} T = \dim W = \dim V \\ &\implies \dim \ker T = 0 \\ &\implies T \text{ is injective.} \end{aligned}$$

Hence T is invertible. □

Theorem 8.4

Let $T: V \rightarrow W$ and $S: W \rightarrow V$ be linear maps with $\dim V = \dim W < \infty$. Then

$$S \circ T = I_V \iff T \circ S = I_W.$$

Proof. ($\implies$) Assume $S \circ T = I_V$. Let $v \in \ker T$. Then

$$v = I_V(v) = (S \circ T)(v) = S(T(v)) = S(0_W) = 0_V.$$

Thus

$$\ker T = \{0_V\}$$

so T is injective.

Since $\dim V = \dim W < \infty$, T is invertible let $T^{-1}: W \rightarrow V$ be its inverse.

Then

$$\begin{aligned} S &= S \circ I_W = S \circ (T \circ T^{-1}) \\ &= (S \circ T) \circ T^{-1} \\ &= I_V \circ T^{-1} = T^{-1}. \end{aligned}$$

Hence

$$T \circ S = T \circ T^{-1} = I_W.$$

($\Leftarrow$) Repeat the proof with T and S swapped, and V and W swapped as well. $\square$

8.3 Isomorphism

Definition 8.2

Let $T: V \rightarrow W$. T is called an *isomorphism* if T is invertible. In this case we write

$$V \cong W,$$

and V and W are called *isomorphic*.

Theorem 8.5

Composition of isomorphisms is an isomorphism.

If $S: U \rightarrow V$ and $T: V \rightarrow W$ are isomorphisms, then $T \circ S: U \rightarrow W$ is an isomorphism.

S, T are isomorphisms.

$$\begin{array}{ccccc} & \xleftarrow{S^{-1}} & & \xleftarrow{T^{-1}} & \\ U & \xrightarrow{S} & V & \xrightarrow{T} & W \\ & \searrow & & \nearrow & \\ & T \circ S & & & \end{array}$$

$T \circ S: U \rightarrow W$ is an isomorphism.

Proof. Since S and T are isomorphisms, there exist

$$S^{-1}: V \rightarrow U, \quad T^{-1}: W \rightarrow V.$$

Claim

$$S^{-1} \circ T^{-1} = (T \circ S)^{-1}.$$

Indeed,

$$\begin{aligned} T \circ S \circ (S^{-1} \circ T^{-1}) &= T \circ \underbrace{(S \circ S^{-1})}_{I_V} \circ T^{-1} \\ &= T \circ I_V \circ T^{-1} \\ &= T \circ T^{-1} \\ &= I_W \end{aligned}$$

and

$$\begin{aligned}
(S^{-1} \circ T^{-1}) \circ (T \circ S) &= S^{-1} \circ \underbrace{(T^{-1} \circ T)}_{I_V} \circ S \\
&= S^{-1} \circ I_V \circ S \\
&= S^{-1} \circ S \\
&= I_U
\end{aligned}$$

Thus $T \circ S$ is invertible with inverse $S^{-1} \circ T^{-1}$:

$$(T \circ S)^{-1} = S^{-1} \circ T^{-1}.$$

Therefore $T \circ S$ is an isomorphism. □

Lemma 8.6

Let V be an n -dimensional vector space over $\mathbb{F}$, and let $\{v_1, \dots, v_n\}$ be a basis of V . Then there exists a unique isomorphism

$$\psi: \mathbb{F}^n \rightarrow V$$

such that

$$\psi(e_i) = v_i \quad (i = 1, \dots, n).$$

Proof. Define linear map

$$\psi(\alpha_1, \dots, \alpha_n) = \alpha_1 v_1 + \dots + \alpha_n v_n.$$

Since

$$\text{span}\{v_1, \dots, v_n\} = V \implies \psi \text{ is surjective}$$

Since

$$\{v_1, \dots, v_n\} \text{ is linearly independent} \implies \ker \psi = \{0\}$$

So ψ is injective. Hence ψ is an isomorphism. □

Theorem 8.7

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$. Then

$$V \cong W \iff \dim V = \dim W.$$

Proof. ($\Rightarrow$) Suppose $V \cong W$, then there exists an invertible linear map $T: V \rightarrow W$, which is injective and surjective.

$$\begin{aligned}
\dim V &= \dim \ker T + \dim \text{Im } T \\
&= 0 + \dim W \\
&= \dim W.
\end{aligned}$$

($\Leftarrow$) Suppose $\dim V = \dim W = n$. Choose basis for V and W

$$\begin{aligned}
&v_1, \dots, v_n \\
&w_1, \dots, w_n
\end{aligned}$$

By the lemma above, there are isomorphisms

$$\begin{aligned}\psi: \mathbb{F}^n &\rightarrow V \\ e_i &\mapsto v_i,\end{aligned}$$

and

$$\begin{aligned}\chi: \mathbb{F}^n &\rightarrow W \\ e_i &\mapsto w_i,\end{aligned}$$

for $i = 1, \dots, n$. Then

$$\chi \circ \psi^{-1}: V \rightarrow W$$

is a composition of isomorphisms, hence an isomorphism.
Therefore

$$V \cong W$$

□

9 Lecture 9

From linear map to the matrices.

9.1 The space of matrices

Definition 9.1

Let V and W be finite-dimensional vector spaces over $\mathbb{F}$.

$$\mathbb{F}^{m,n} = \{m \times n \text{ matrices over } \mathbb{F}\}.$$

$(m \times n)$ matrices m #rows n #columns.

$\mathbb{F}^{m,n}$ is a vector space over $\mathbb{F}$.

$$[A_{ij}]_{\substack{i=1,\dots,m \\ j=1,\dots,n}} = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}$$

$$\begin{aligned}[A_{ij}] + [B_{ij}] &= [A_{ij} + B_{ij}] \\ \lambda[A_{ij}] &= [\lambda A_{ij}]\end{aligned}$$

Basis of $\mathbb{F}^{m,n}$ consists of matrices

$$E_{i=1,\dots,m \atop j=1,\dots,n}$$

With a single entry 1 at the (i, j) -th position and the rest 0.

$$E_{ij} = \begin{bmatrix} & & j & & \\ 0 & \cdots & & \cdots & 0 \\ \vdots & & & & \vdots \\ & \cdots & 1 & \cdots & \\ \vdots & & & & \vdots \\ 0 & \cdots & & \cdots & 0 \end{bmatrix} i$$

$$A = \sum_{i=1}^m \sum_{j=1}^n A_{ij} E_{ij}.$$

So the dimension of $\mathbb{F}^{m,n}$ is the number of different place in the matrix, which is mn .

$$\dim \mathbb{F}^{m,n} = mn$$

9.2 Matrix representation of linear maps

Definition 9.2

Matrix multiplication

$$\mathbb{F}^{m,n} \times \mathbb{F}^{n,p} \rightarrow \mathbb{F}^{m,p}$$

$$[A_{ij}] \times [B_{jk}] \mapsto [C_{ik}]$$

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots \\ A_{m1} & A_{m2} & A_{m3} & \cdots \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} & B_{13} & \cdots \\ \vdots & \ddots & \vdots & \ddots \\ B_{n1} & B_{n2} & B_{n3} & \cdots \end{bmatrix}$$

$$C_{11} = A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} + \cdots + A_{1n}B_{n1}$$

Theorem 9.1

$A \in \mathbb{F}^{m,n} \sim$ linear map $\mathbb{F}^n \rightarrow \mathbb{F}^m$

$$\mathbb{F}^n \rightarrow \mathbb{F}^m$$

$$x \mapsto Ax$$

$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}$$

For vector spaces V and W

$$\dim V = n \quad \dim W = m$$

T is a linear map from V to W

$$V \xrightarrow{T} W$$

Choose basis

$$\{v_1, \dots, v_n\} \quad \text{for } V$$

$$\{w_1, \dots, w_m\} \quad \text{for } W$$

$$\mathbb{F}^n \xrightarrow[\psi]{\cong} V \quad \mathbb{F}^m \xrightarrow[\psi]{\cong} W$$

$$e_i \mapsto v_i \quad e_i \mapsto w_i.$$

The matrix representation of the linear map T with respect to the chosen bases is given by the following commutative diagram

$$\begin{array}{ccccc} \mathbb{F}^n & \xrightarrow[\psi]{\cong} & V & \xrightarrow{T} & W & \xrightarrow[\psi^{-1}]{\cong} & \mathbb{F}^m \\ & \searrow & & & & \nearrow & \\ & & \psi^{-1} \circ T \circ \psi & & & & \end{array}$$

$$T(v_i) = \sum_{j=1}^m w_j A_{ji}$$

$$\begin{aligned} T[v_1 \ \cdots \ v_n] &= [Tv_1 \ \cdots \ Tv_n] \\ &= [w_1 \ \cdots \ w_m] \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} T\left(\sum_{i=1}^n x_i v_i\right) &= \sum_{i=1}^n x_i T v_i \\ &= \sum_{i=1}^n x_i \sum_{j=1}^m A_{ji} w_j \\ &= \sum_{j=1}^m w_j \left(\sum_{i=1}^n A_{ji} x_i\right) \end{aligned}$$

$\mathcal{M}(T)$ matrix which represents the linear map T with respect to the chosen bases.

Definition 9.3

The coordinate vector of a vector $w \in W$ with respect to the basis $\{w_1, \dots, w_m\}$ is defined as follows:

$$\begin{aligned} \psi^{-1}: W &\rightarrow \mathbb{F}^n \\ w = \sum_{i=1}^m x_i w_i &\mapsto [x_1 \ \cdots \ x_m] \end{aligned}$$

Coordinate vector of w with respect to the basis $\{w_1, \dots, w_m\}$.

Example

$$\mathcal{D}: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$$

$\{1, x, x^2, x^3\}$ basis of $\mathcal{P}_3(\mathbb{R})$ $\{1, x, x^2\}$ basis of $\mathcal{P}_2(\mathbb{R})$ Question: What is the matrix representation of the differentiation operator $\mathcal{D}$ with respect to the given bases?

$$\mathcal{D}(1, x, x^2, x^3) = (0, 1, 2x, 3x^2) = (1, x, x^2) \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$[\mathcal{M}(\mathcal{D})] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

9.3 The isomorphism $\mathcal{L}(V, W) \cong \mathbb{F}^{m \times n}$ and composition

Theorem 9.2

For vector spaces V and W , let $\dim V = n$, $\dim W = m$.

$$\begin{aligned}\mathcal{L}(V, W) &\rightarrow \mathbb{F}^{m \times n} \\ T &\mapsto \mathcal{M}(T)\end{aligned}$$

is a vector space isomorphism, and hence

$$\begin{aligned}\dim \mathcal{L}(V, W) &= mn \\ &= (\dim W)(\dim V)\end{aligned}$$

Proof. (From workshop $\mathcal{M}(T) = \psi^{-1} \circ T \circ \psi$)

$$\begin{aligned}\mathcal{M}(S + T) &= \psi^{-1} \circ (S + T) \circ \psi \\ &= \psi^{-1} \circ S \circ \psi + \psi^{-1} \circ T \circ \psi \\ &= \mathcal{M}(S) + \mathcal{M}(T)\end{aligned}$$

For $x \in \mathbb{F}^n$,

$$\begin{aligned}\psi^{-1}(\underbrace{(S + T)(\psi(x))}_{S(\psi(x)) + T(\psi(x))}) &= \psi^{-1}(S(\psi(x)) + T(\psi(x))) \\ &= \psi^{-1}(S(\psi(x))) + \psi^{-1}(T(\psi(x))) \\ &= (\psi^{-1}S\psi + \psi^{-1}T\psi)(x) \\ \mathcal{M}(\lambda T) &= \psi^{-1} \circ (\lambda T) \circ \psi \\ &= \lambda(\psi^{-1} \circ T \circ \psi) \\ &= \lambda \mathcal{M}(T)\end{aligned}$$

$\mathcal{M}$ is a linear map.

$$\begin{aligned}\mathcal{M}(T) &= \psi^{-1} \circ T \circ \psi \\ \psi \circ \mathcal{M}(T) \circ \psi^{-1} &= \underline{\psi \circ \psi^{-1}} \circ T \circ \underline{\psi \circ \psi^{-1}} \\ &= T\end{aligned}$$

□

Theorem 9.3

Composition of linear maps is represented by matrix multiplication with respect to the chosen bases.

Proof. U, V, W are vector spaces over the field $\mathbb{F}$.

$$\dim U = p, \quad \dim V = n, \quad \dim W = m.$$

Let $S: U \rightarrow V$ and $T: W \rightarrow U$ be linear maps with chosen bases for V, W, U . Then the composition $S \circ T: W \rightarrow V$ is represented as follows

$$\begin{array}{ccccc} W & \xrightarrow{T} & U & \xrightarrow{S} & V \\ & \searrow & & \nearrow & \\ & & S \circ T & & \end{array}$$

Chosen basis

$$U: u_1, \dots, u_p$$

$$W: w_1, \dots, w_m$$

$$V: v_1, \dots, v_n$$

$$Su_i = \sum_{j=1}^n v_j S_{ji} \quad Tw_k = \sum_{i=1}^p u_i T_{ik}$$

$$(S \circ T)w_k = \sum_{j=1}^n v_j C_{jk}$$

where

$$C_{jk} = \sum_{i=1}^p S_{ji} T_{ik} \Leftrightarrow \mathcal{M}(S \circ T) = \mathcal{M}(S)\mathcal{M}(T)$$

Then

$$\begin{aligned} S(Tw_k) &= S\left(\sum_{i=1}^p u_i T_{ik}\right) \\ &= \sum_{i=1}^p T_{ik} S(u_i) \\ &= \sum_{i=1}^p T_{ik} \left(\sum_{j=1}^n v_j S_{ji}\right) \\ &= \sum_{j=1}^n v_j \left(\sum_{i=1}^p S_{ji} T_{ik}\right) \\ &= \sum_{j=1}^n v_j C_{jk} \end{aligned}$$

□

9.4 Normal Form for Matrices

Normal form for matrices representing linear maps (cf. proof of the Rank theorem)

For linear map T

$$T: V \rightarrow W$$

Choose basis for $\ker T$

$$u_1, \dots, u_k$$

Complete it to a basis for the whole space V

$$v_1, \dots, v_r, u_1, \dots, u_k$$

So basis for the $\operatorname{im} T \subseteq W$ is

$$Tv_1, \dots, Tv_r$$

Complete it to a basis for the space W

$$Tv_1, \dots, Tv_r, w_1, \dots, w_p$$

apply the linear map T to the basis vectors of V

$$\begin{aligned} T(v_1, \dots, v_r, u_1, \dots, u_k) &= (Tv_1, \dots, Tv_r, Tu_1, \dots, Tu_k) \\ &= (Tv_1, \dots, Tv_r, 0, \dots, 0) \\ &= (Tv_1, \dots, Tv_r, w_1, \dots, w_p) \end{aligned}$$

Since

$$\forall i = 1, \dots, k \quad Tu_i = 0$$

So if we choose the bases for V and W as described above, the matrix representation of T with respect to these bases is, which called the *normal form* of the matrix representation of T .

$$\mathcal{M}(T) = \left[\begin{array}{ccc|c} & r & & n \\ 1 & \cdots & 0 & \\ \vdots & \ddots & \vdots & 0 \\ 0 & \cdots & 0 & \\ \hline & & 0 & 0 \\ & & & p \end{array} \right] \begin{matrix} r \\ p \end{matrix}$$

$$r = \text{rank } T = \dim \text{im } T$$

$$\mathcal{M}(T) = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right)$$

Remark

Careat: For a linear map

$$T: V \rightarrow W$$

The problem will be much harder.

cf. Jordan Normal Form